

CYFI750 2025 Midterm (100 Points)

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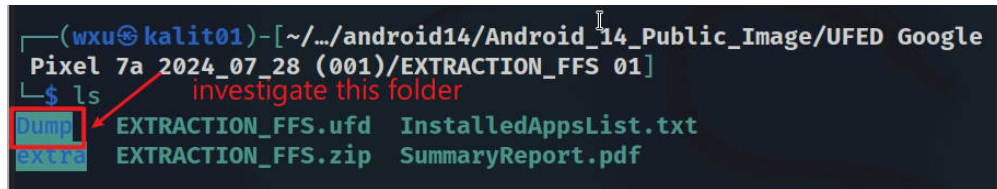
Instructions

- **Format & Scoring:** This is a timed, open-book exam. Each question is worth 20 points. Full credit is awarded only for correct results supported by documented evidence.
- **Technical Setup:** You are responsible for testing both Kali environments prior to the exam. Conduct all tasks via the **Linux Terminal**.
- **Academic Integrity:** Collaboration, sharing results, and discussion are strictly prohibited.
- **Command Restrictions:** You **MUST** only use commands covered in the course curriculum. If a task requires a command not introduced in class, you must provide a written **justification** of why commands covered in the lectures can't solve the problem; otherwise, no credit will be awarded.
- **Evidence & Investigation:** Answers submitted without documented investigation processes and supporting evidence will receive zero points. You must **highlight** your final answers. If you believe the provided image is **corrupted** or does not contain the answer, you must provide proof to support your claim.
- **Submission:** The Canvas Dropbox closes automatically before the deadline. Submit your work several minutes early to account for potential discrepancies between your local clock and the Canvas server time.
- **Documentation:** Refer to the provided guidelines for specific instructions regarding required screenshots. https://github.com/frankwxu/digital-forensics-lab-p2/blob/main/practice/Assignment_Instructions.pdf

1 Obtain Android 14 image (practice before the exam)

- a. There is a compressed file `Android.14.Public.Image.tar.gz` (already downloaded and placed under the folder `/lab/android14`). Alternatively, you can download the compressed file here. Verify the hash code (MD5) of the compressed file
`Android.14.Public.Image.tar.gz` is 2F9578715A315C0897E51EF9C1007F2D.

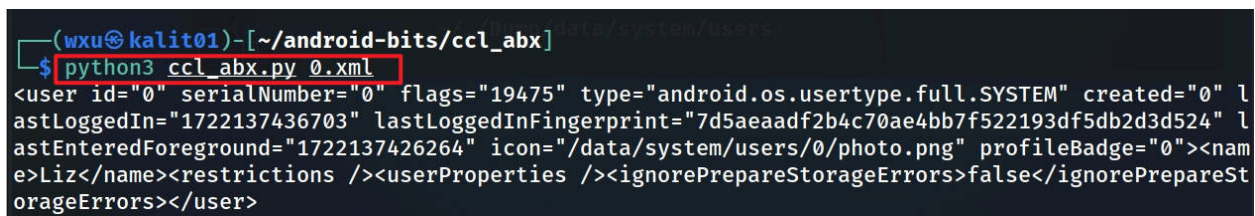
- b. The .gz file contains a compressed Android disk image (logic files) in a folder **dump** shown in Figure 1. Note that you must uncompress (using **tar** or other tools) the logic image to answer the rest of the questions. No information should be used outside of the **Dump** folder.



A terminal window showing the command `ls` being executed in the directory `~/android14/Android_14_Public_Image/UFED Google Pixel 7a 2024_07_28 (001)/EXTRACTION_FFS 01`. The output lists four files: `Dump`, `EXTRACTION_FFS.ufd`, `InstalledAppsList.txt`, `EXTRACTION_FFS.zip`, and `SummaryReport.pdf`. The `Dump` folder is highlighted with a red box, and a red arrow points to it with the text "investigate this folder".

Figure 1: The **Dump** folder you need to investigate.

- c. You may need to change the permission of the **Dump** folder (e.g., `chmod -R`) to access all folders. Limit your investigation activities inside the **Dump** folder.
- d. Android 12+ uses binary XML instead of XML. You need to download a tool using an open source tool from GitHub with the command
`git clone https://github.com/cclgrouppltd/android-bits.git`,
and use the following command `python` or `Python3` to convert binary XML to XML format shown in Figure 2.



A terminal window showing the command `python3 ccl_abx.py 0.xml` being executed in the directory `~/android-bits/ccl_abx`. The output is a valid XML snippet representing an Android user profile.

```
<user id="0" serialNumber="0" flags="19475" type="android.os.usertype.full.SYSTEM" created="0" lastLoggedIn="1722137436703" lastLoggedInFingerprint="7d5aeaadf2b4c70ae4bb7f522193df5db2d3d524" lastEnteredForeground="1722137426264" icon="/data/system/users/0/photo.png" profileBadge="0"><name>Liz</name><restrictions /><userProperties /><ignorePrepareStorageErrors>false</ignorePrepareStorageErrors></user>
```

Figure 2: Using Python code to convert a file `0.xml` in Binary XML to XML.